

# **Improved performance and stability in quantum dot solar cells through band alignment engineering**

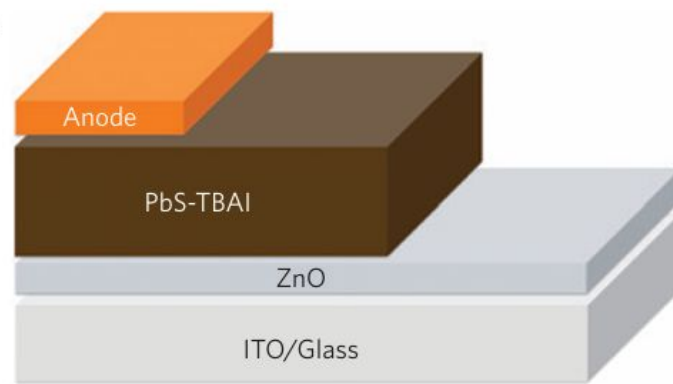
Chia-Hao M. Chuang, Patrick R. Brown, Vladimir Bulovic and Mounqi G. Bawendi, Nat. Mater., Vol 13, 796-801 (2014)

**Camille De Le Vallée & Ken Zou**  
**16 février 2026**

# Context of the study

Problems of the previous solar cells:

- poor air stability
- inert-atmosphere processing environment or high temperature processing



**Development of room-temperature solution-processed ZnO/PbS quantum dot solar cells + band alignment with  $\pi$  ligands**

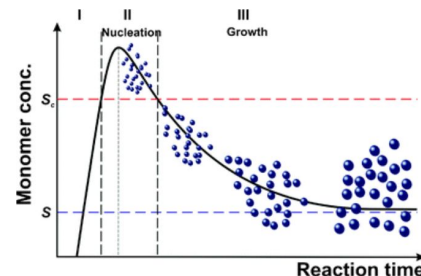
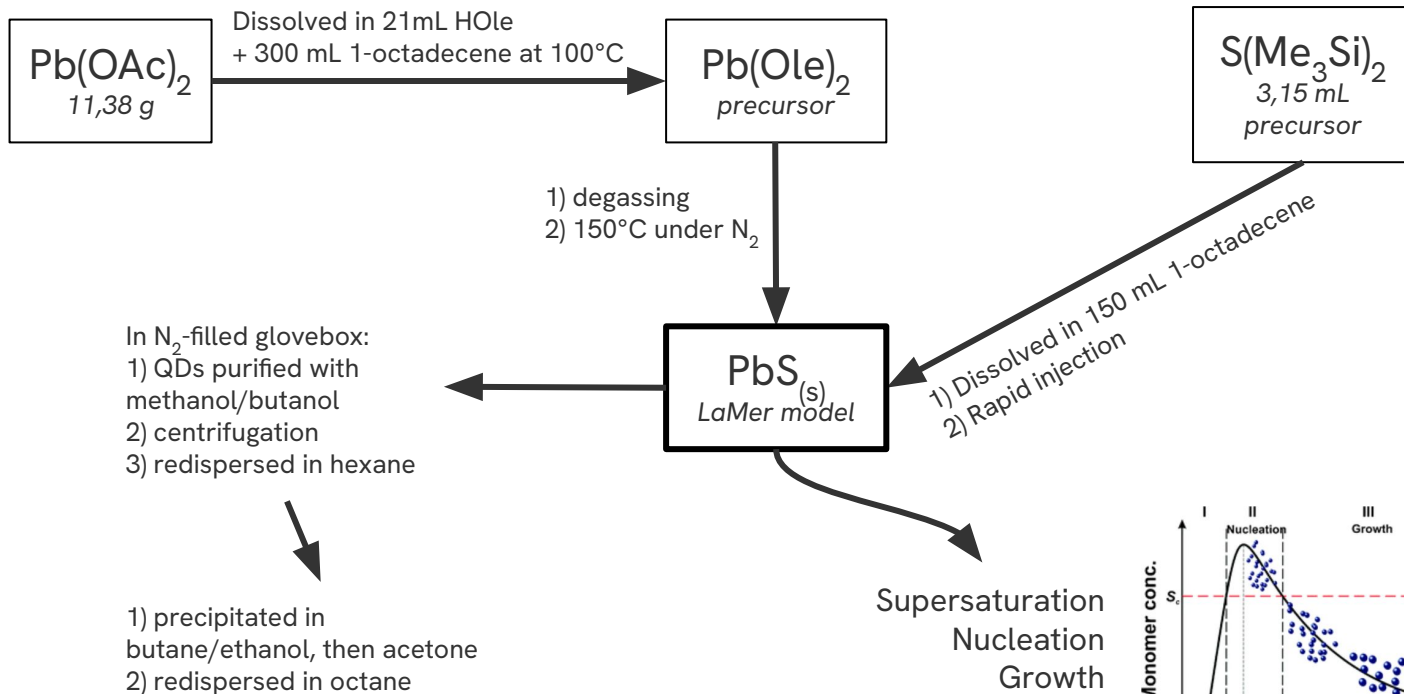
PbS QDs absorb photons  $\rightarrow$  electron-hole pairs  $\rightarrow$  charge separation at ZnO/PbS interface  $\rightarrow$  charges collected

**02**

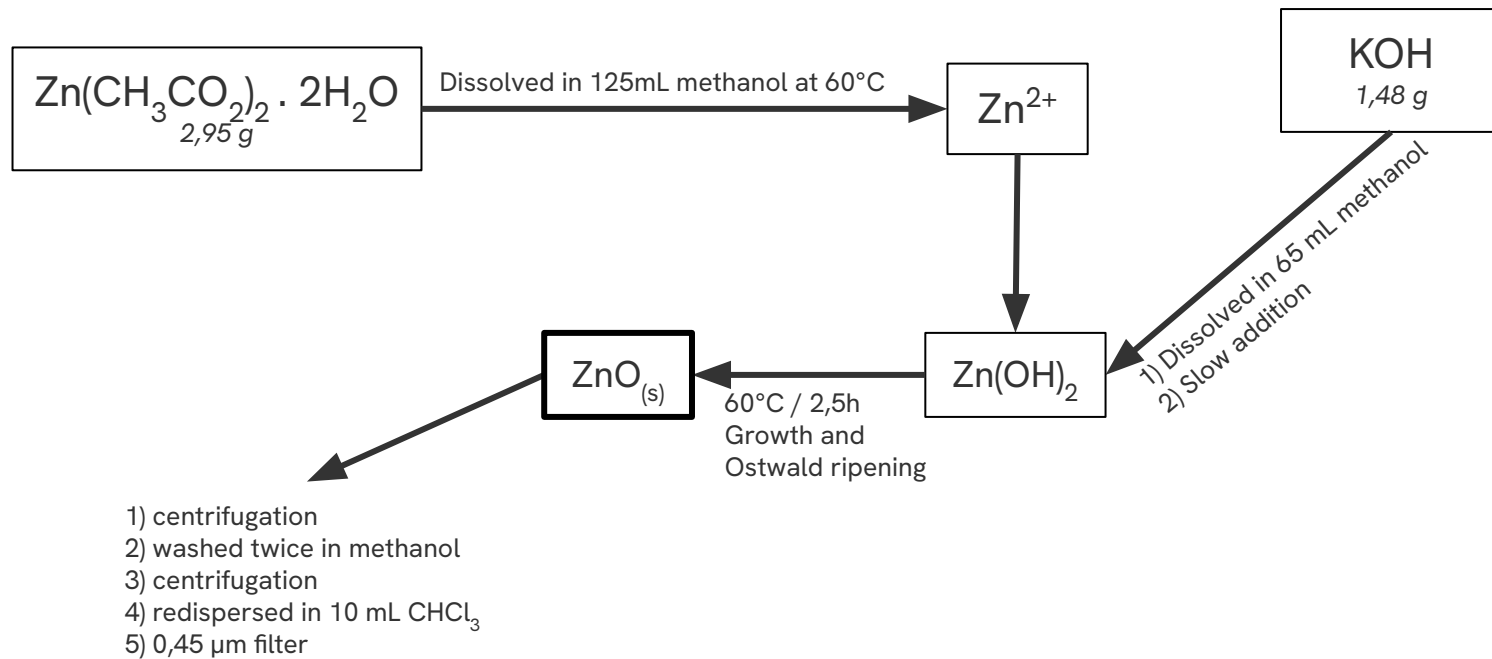
# **Synthetic procedures**

PbS QDs, ZnO nanoparticles and device architecture

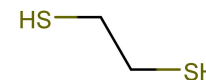
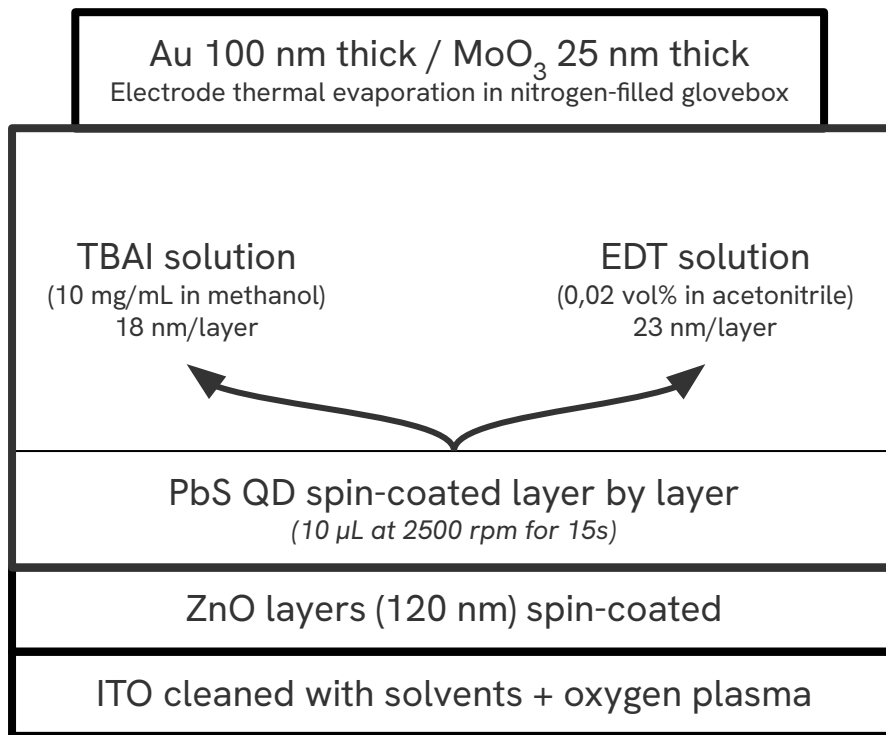
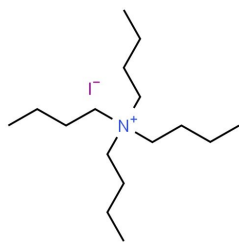
# Synthesis of colloidal PbS QDs



# Synthesis of ZnO nanoparticles



# Device fabrication

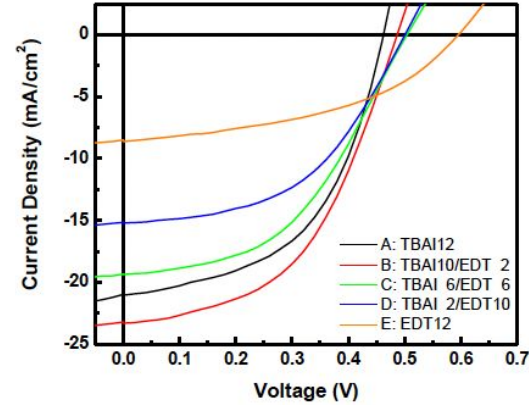
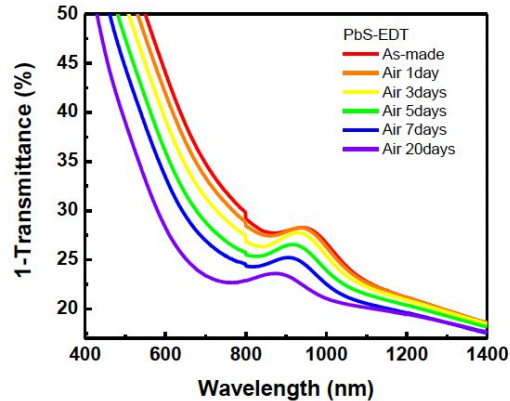
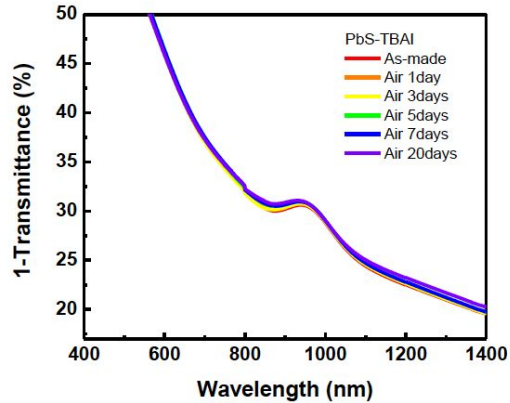


**03**

# **Main results**

Power conversion efficiency and air stability

# Comparison of device performances



PbS-TBAI (compared to PbS-EDT):

- better air stability
- higher current density

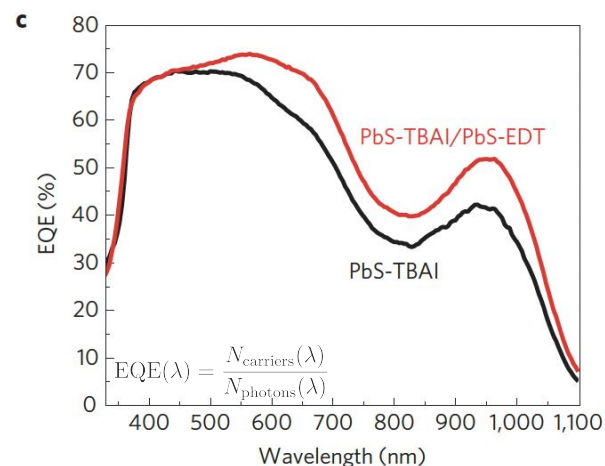
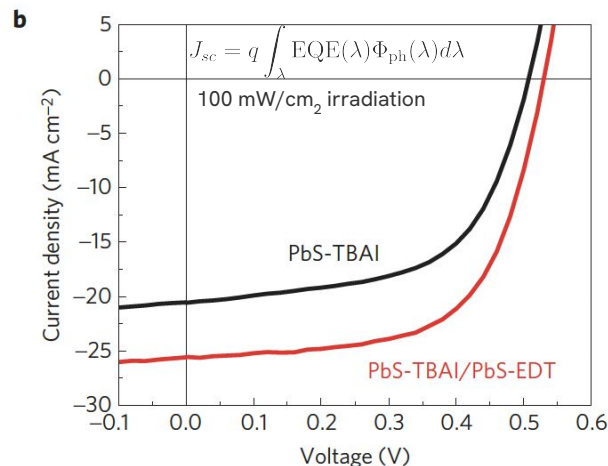
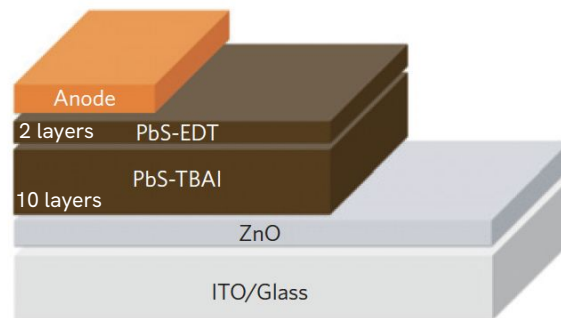
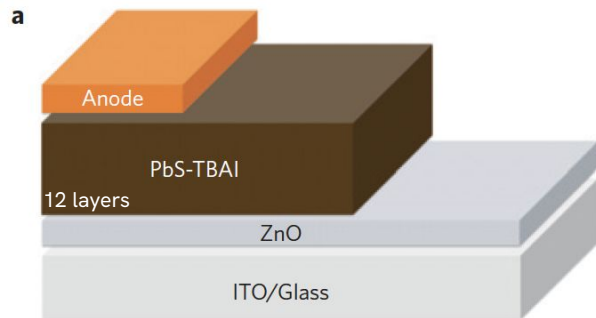


# Comparison of device performances

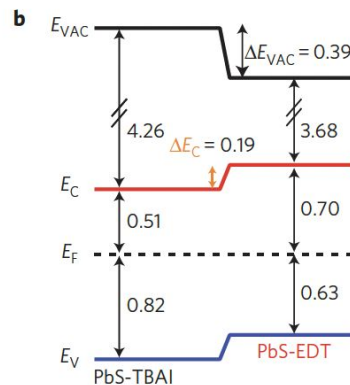
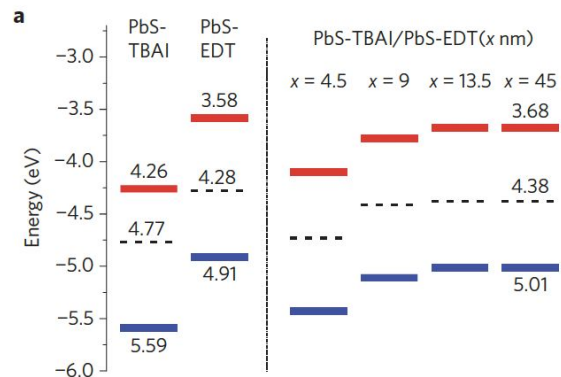
But...

**20% improvement in  $J_{sc}$**   
From  $20.7 \pm 1.1$  to  $25.3 \pm 1.1 \text{ mA/cm}^2$

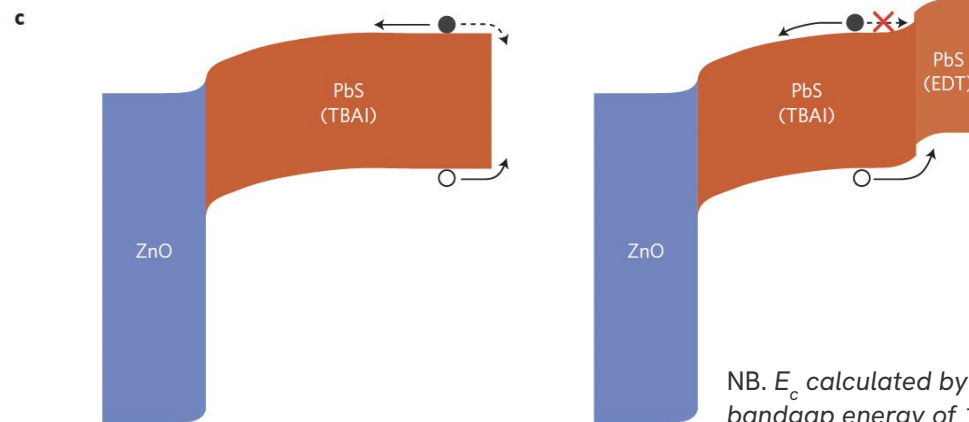
**35% improvement in PCE**  
From  $6.0 \pm 0.4\%$  to  $8.2 \pm 0.6\%$



# Performance improved with band offset



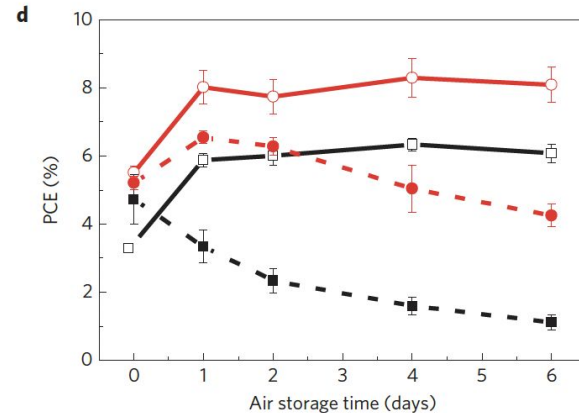
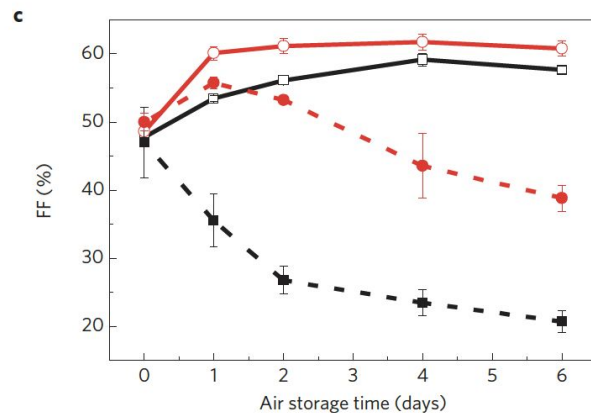
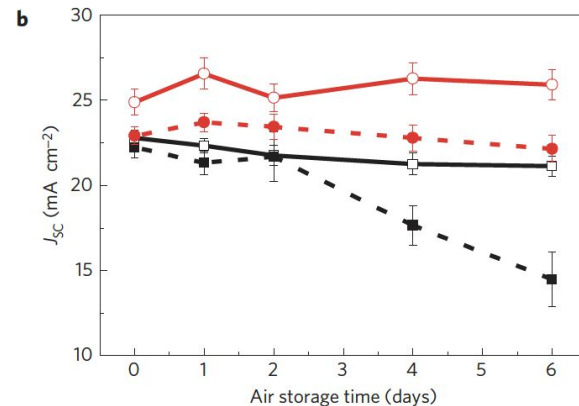
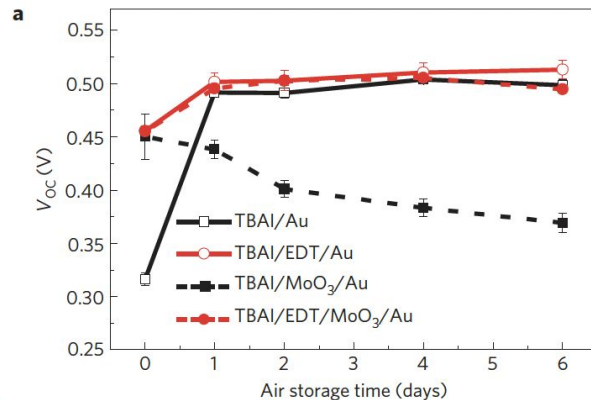
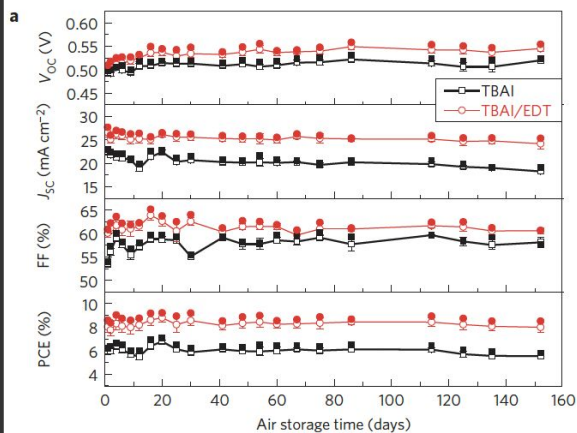
Ultraviolet Photoelectron Spectroscopy (UPS)



- No backflow of electrons
- Less interfacial recombination
- Higher EQE at longer wavelengths (depletion junction at the interface)

NB.  $E_c$  calculated by adding the optical bandgap energy of 1.33 eV (930 nm).

# Device stability with anodes



- S-shape in J-V : Schottky barrier
- MoO<sub>3</sub> can be reduced, hydrated and adsorb if exposed to air
- MoO<sub>3</sub> is selective

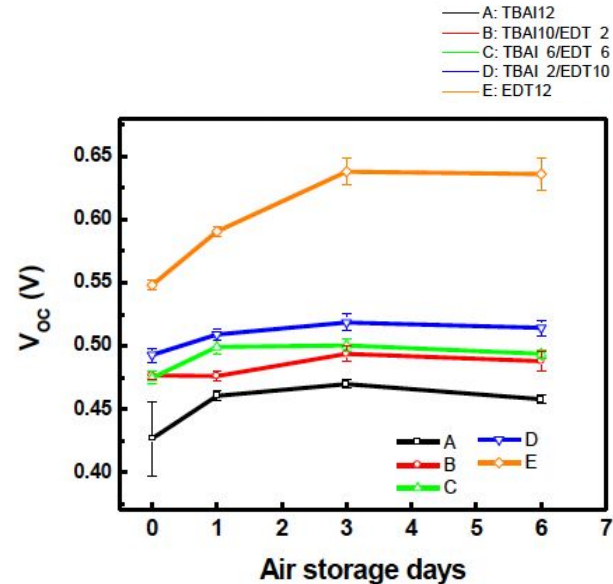
## Major findings

- Demonstrated high-performance QD solar cells through the engineering of band alignment at QD/QD and QD/anode interfaces.
- Solar cells are solution-processed in air at room temperature.
- Good air stability with Au anodes and TBAI/EDT 10/2 ratio.

## Major challenge

- Need to increase the open-circuit voltage because they only got

$$qV_{oc} < \frac{E_g}{2}$$



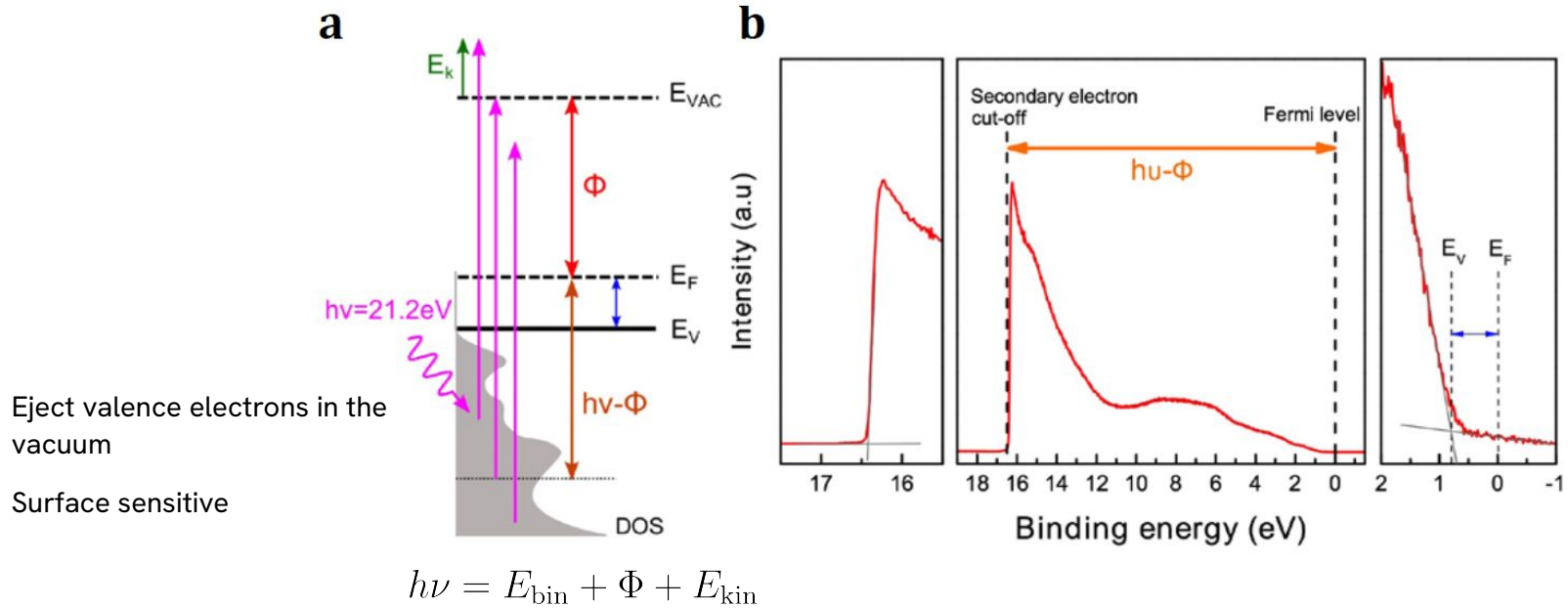


# **Annexes**

Additional information about the article

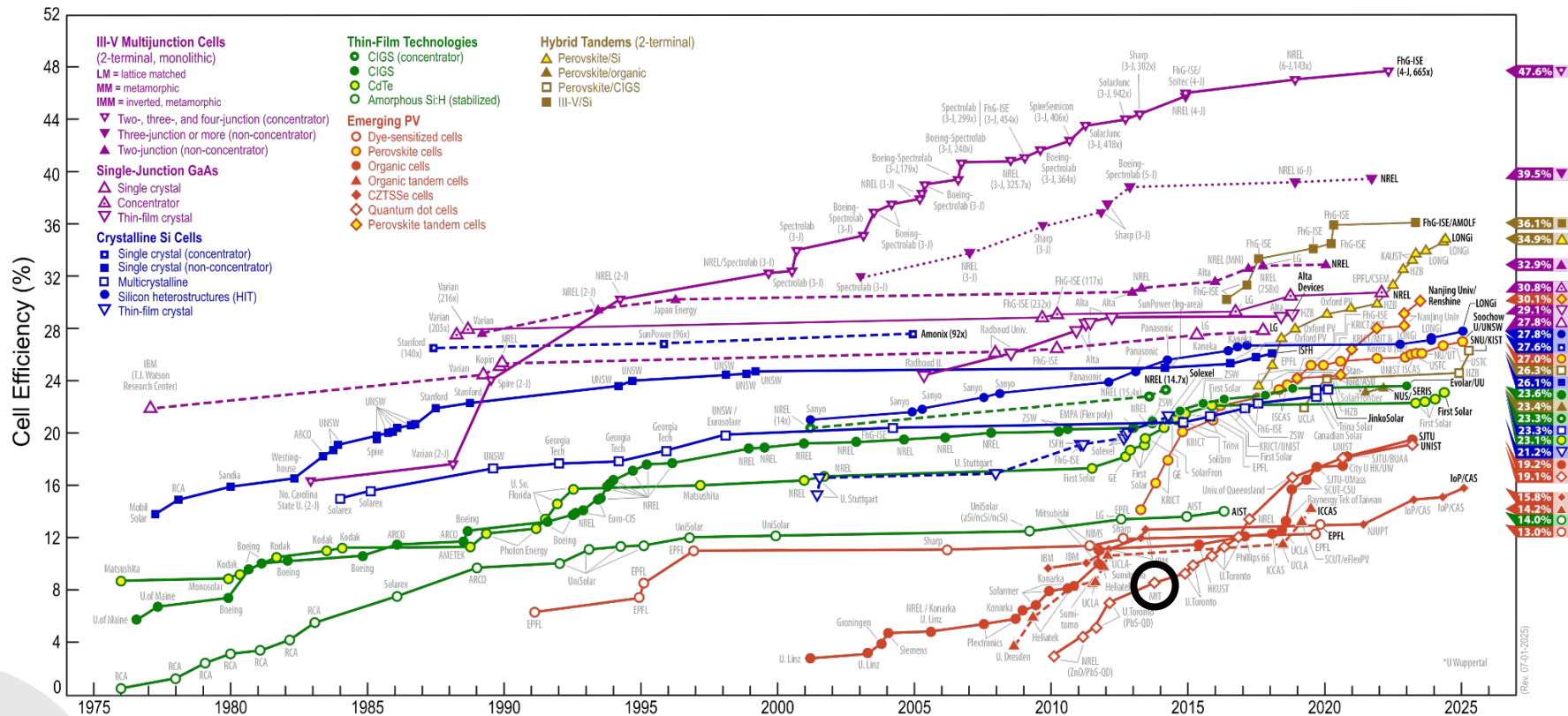


# Ultraviolet Photoelectron Spectroscopy (UPS)

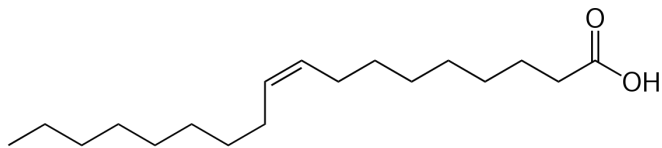


**Figure S4 | Basic principles of UPS measurements.** **a**, Schematic illustration of the energy diagram. DOS: density of states in the valence band. **b**, An example UPS spectrum. The left panel shows the magnified view of the high binding energy region and the right panel shows the magnified view of the low binding energy region. Grey lines represent the linear fits.

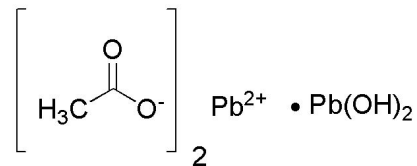
# Best Research-Cell Efficiencies



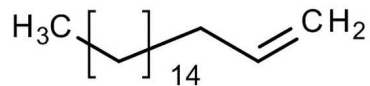
# Molecule library



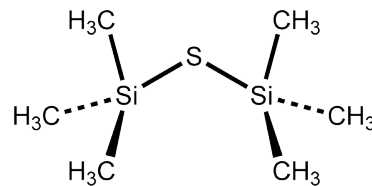
Oleic acid



Lead acetate

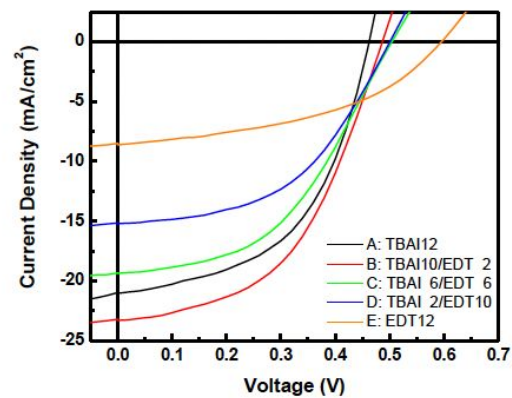
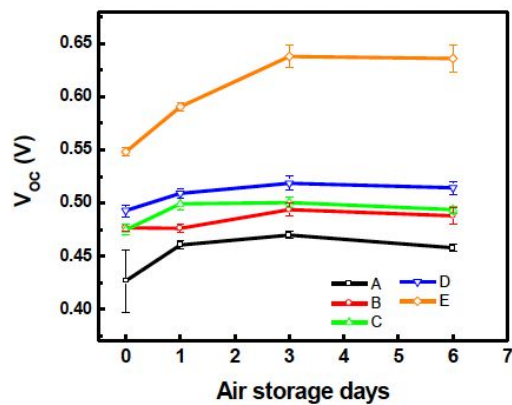


1-octadecene



Hexamethyldisilathiane



**a****b****c**